# CCCA Cell-State Analysis — Ledergor et al. 2018

Multiple Myeloma · MARS-seq · 8,182 malignant plasma cells · 16 samples

## GO / NO-GO GATES

### Gate 1 — mixing & continuity

**GO**

Harmony (θ=15) fuses the patient-private myeloma clones: same-sample neighbour fraction 0.87→0.26, iLISI 1.0→5.1 / 16, one connected kNN component. Judged on the regenerated embedding across a θ-sweep — samples intermix on one continuous, uniformly low-cycling body (no proliferative lobe to scope).

### Gate 2 — module geography

**GO (non-branching)**

Signatures collapse onto a single (Gast/pSom)→Crest axis: smoothed pSom↔Crest r=−0.97, Gast↔pSom r≈−0.10 (no cross-repression → no two competing arms). Crest is an inverse terminal gradient (crest-high decile pseudotime 0.83; joint pSom&Crest enrichment 0.16). Principal curve is fold-free (2 tips / 0 forks).

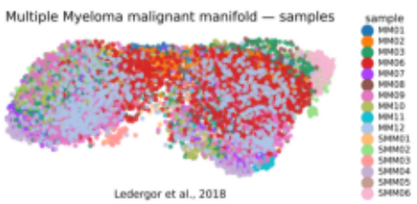
### Gate 3 — hub cross-repression

**NA**

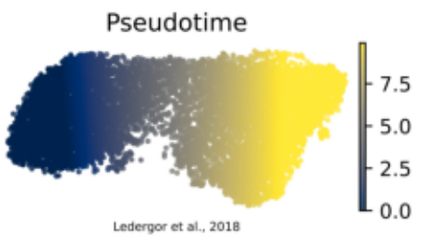
Non-branching topology — no bifurcation and no Gast/pSom hub, so the branching-only cross-repression test does not apply. Correlation & modules notebooks are correctly not produced.

## STEP FIGURES

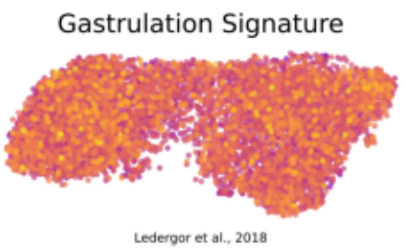
01 · Samples (integration)



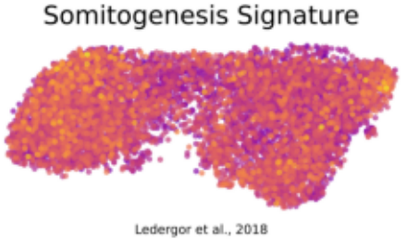

02 · Pseudotime (curve)



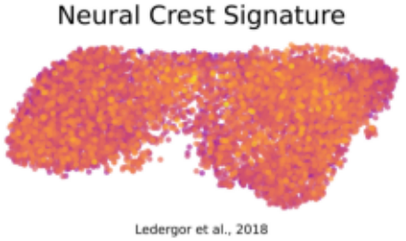

03 · Gastrulation



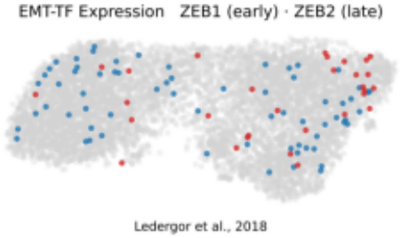

03 · Somitogenesis




03 · Neural Crest




04 · EMT-TFs (ZEB1→ZEB2)




**Dataset:** Multiple myeloma (Ledergor 2018), MARS-seq, 8,182 malignant plasma cells over 16 samples.

**Verdict:** Pattern reproduces as the NON-BRANCHING case — a single gastrulation/somitogenesis→neural-crest developmental axis; no hub, no bifurcation.

## PARAMETER CHOICES & RATIONALE

| | |
|---|---|
| **Min-cell threshold** | ≥95 — geometric mean of the 183→50 spillover cliff (16/18 samples; drops MM04=26, MM05=50) |
| **Integration** | Harmony on sample, θ=15 (strong; patient-private MM clones), n_pcs=40, n_neighbors=30, spread=1.3 |
| **Islands removed** | 15 cells / 2 detached specks (UMAP radius graph, largest component) |
| **Geometry** | Non-branching → scFates principal curve (ElPiGraph), Nodes=20, λ=0.3, μ=100; 2 tips / 0 forks, fold-free |
| **Root node** | Curve tip 0 — Gast/pSom-high, Crest-low (developmental origin; pSom→Crest along pseudotime) |
| **Module means (raw)** | Gast −0.029 · pSom +0.016 · Crest +0.004 → Linear-pSom/Crest |
| **EMT-TFs shown** | ZEB1 (root) → ZEB2 (terminus) — only detectably-expressed factors (ZEB1 3.3%, ZEB2 1.5%) |